

CHATPTER AND SYLLABUS

Syllabus of LA

Chapter :1

1. 1 ,1.2 ,1.3 ,1.4 , 1.5 , 1.6 , 1.7

Chapter :2

2.1 , 2.2 ,2.3

Chapter :3

3.1, 3.2 ,3.3 , 3.4 , 3.5

Chapter :4

4.1 4.2 4.3 4.4 4.5 4.6 4.7 4.8.4.9

Chapter :5

5.1 - 5.2

Chapter :6

6.1 , 6.2 , 6.3

Chapter :8

8.1 , 8.2 , 8.3 , 8.4 , 8.5

Important note:

These are practice questions uploaded by my teacher on the google classroom and some are given by seniors , i want to give more priority to ,my teacher's questions

So you will solve first my teachers questions , and then the one by seniors , proper mention the headings teacher and senior and , and then if some questions are common you will redo them no matter what , we will only do EXERCISES as mentioned in this

Practice questions as uploaded by my teacher :

Chapter 1:

1.1	Q 19
1.2	Q 13
1.3	Q 15, 34
1.4	Q 1, 16, 25
1.5	Q 4, 21
1.6	Q 10, example 4
1.7	Q 10, 25, 27

Practice Questions

2.1

Q 31,
33, 39, 41

2.2

Q 3, 5,
11, 21, 25, 27, 29

2.3

Q 5, 13, 17, 21, 27

Chapter 3: Euclidean Vector Spaces

Ex # 3.1

Questions: 7, 18

Ex # 3.2

Questions: 5, 8, 10, 12, 14, 18, 26

Ex # 3.3

Questions: 2, 4, 8, 10, 11, 14, 17, 19, 22, 26, 28, 33, 36, 37.

Ex # 3.4

Questions: 4, 5, 7, 10, 12, 15, 17, 22, 24, 25, 28, 29.

Ex # 3.5

Questions: 1, 5, 10, 13, 17, 21, 25, 28, 31, 32.

Chapter 4

Ex. 4.1 Q# 5, 12

Ex. 4.2 Q# 4, 11, 21

Ex.4.2

Q# 1(b), 2(c), 3(a), 7, 9, 16(a)

Ex 4.3 Q# (2a, 10, 15)

Ex 4.4 Q# (5b, 11)

Ex 4.5 Q# (8, 13b)

Ex. 4.6 Q# 2, 7, 9, 13

Ex. 4.8 Q# 2, 5, 8, 9

Ex. 4.9 Q# 1, 5

CHAPTER 5

SECTION 5.1: Eigenvalues and eigenvectors and eigenspaces, related examples.

EXERCISE 5.1 Questions 1, 6, 10, 16, 22, 27

SECTION 5.2: Similar matrices, Diagonalization, related examples.

EXERCISE 5.2: Questions 1, 5, 8 10, 15, 20, 24 30, 31, 34.

Linear Algebra

CHAPTER 6

SECTION 6.1: Definition of inner product. Various types of inner products defined on Euclidean space, space of all matrices, space of all polynomials and space of all continuous functions. Length of

a vector and distance between two vectors using inner product.

EXERCISE 6.1 Questions 1-28

SECTION 6.2: Angle between two vectors, orthogonality criteria, related examples. Cauchy Schwarz

inequality (application only)

EXERCISE 6.2: Questions 1-24.

Section 6.3: Orthogonal and orthonormal sets, orthogonal and orthonormal bases, the Gram Schmidt Process of orthonormalization

Exercise 6.3: Q 1-14, Q 27-31

Chapter 8:

Practice questions given by seniors :

Chapter 1

Exercise 1.1

Q3(b, c), Q4(c), Q9(b, e), Q11(a, b, c), Q13(a, c), Q15(a), Q17(a)

Exercise 1.2

Q1(c, g), Q3(c, d), Q4(c), Q7, Q8, Q9, Q13, Q14, Q21, Q22, Q25, Q36

Exercise 1.3

Q1(c, f), Q2(d, f), Q3(d, e), Q5(g, i), Q11(a, b), Q14(a), Q15, Q23, Q24, Q36

Exercise 1.4

Q2(b, d), Q5, Q7, Q10, Q13, Q17, Q21(b), Q25, Q43(a), Q45(a, b)

Exercise 1.5

Q2(d), Q4(c, d), Q6(b, c), Q7, Q8, Q9(a), Q11(1), Q12(b), Q10(b), Q13, Q21, Q23, Q27

Exercise 1.6

Q1, Q3, Q9, Q11(ii), Q15, Q19

Exercise 1.7

Q1(c), Q5, Q9, Q12, Q14, Q19, Q21, Q26, Q28, Q32

Chapter 2

Exercise 2.1

Q1, Q3(d), Q7, Q9, Q13, Q14, Q15, Q18, Q20(a, f), Q23, Q25, Q31, Q33(b), Q35

Exercise 2.2

Q3, Q5, Q6, Q9, Q12, Q14, Q17, Q22, Q23, Q25, Q27, Q29

Exercise 2.3

Q1, Q3, Q6, Q7, Q11, Q14, Q15, Q18, Q19, Q23, Q25, Q26, Q28, Q31

Supplementary Exercise

Q14, Q16, Q25

Chapter 3

Exercise 3.1

Q6(a), Q8(b), Q9(a), Q15(c), Q16, Q18, Q20, Q21

Exercise 3.2

Q1(b), Q4, Q7, Q10, Q11(b), Q15(d), Q17(b)

Exercise 3.3

Q3, Q9, Q11, Q13(b), Q15, Q19, Q21, Q25, Q28, Q29, Q30

Exercise 3.4

Q4, Q6, Q8, Q9, Q12, Q13, Q15, Q18

Exercise 3.5

Q7, Q9, Q10, Q12, Q13, Q15, Q17, Q19, Q21, Q26(a), Q39

Chapter 4

Exercise 4.1

Q2, Q8, Q10, Q12

Exercise 4.2

Q2, Q5(b), Q10, Q12(b), Q13(b), Q18

Exercise 4.3

Q4(a), Q6(a), Q8(d), Q9, Q15, Q20

Exercise 4.4

Q2(a), Q5(b), Q7(a), Q9, Q10(b), Q16(a), Q19

Exercise 4.5

Q2, Q5, Q7(b), Q12(a), Q13(b), Q15, Q18, Q21(b), Q22

Exercise 4.6

Q3, Q5, Q7(d), Q10, Q11(a), Q13, Q18, Q27

Exercise 4.7

Q1, Q3, Q4, Q6, Q7, Q9, Q16

Exercise 4.8

Q1(b), Q8(a), Q10(a), Q11(a), Q13, Q15, Q17

Chapter 5

Exercise 5.1

Q3, Q6(d), Q9, Q12, Q14, Q15

Exercise 5.2

Q3, Q7, Q9, Q11, Q16, Q18, Q19, Q30, Q33

Chapter 6

Exercise 6.1

Q1(f), Q5, Q7, Q11, Q13, Q15, Q22, Q26, Q27

Exercise 6.2

Q2(b), Q4, Q5, Q7(b), Q9, Q11, Q17

Exercise 6.3

Q2(b), Q4(a), Q6, Q10, Q23, Q29, Q31, Q37, Q45, Q47, Q48